

Windows PowerShell Laboratory Manual

Basic Lab Exercises for BCA.

Experiment 1: Exploring PowerShell

Aim:

Learn the associated PowerShell cmdlets.

Commands:

Get-Host

Get-Location

Get-Date

Get-Command

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 2: Help System

Aim:

Learn the associated PowerShell cmdlets.

Commands:

Get-Help Get-Process

Get-Help Get-Service -Examples

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 3: Processes

Aim:

Learn the associated PowerShell cmdlets.

Commands:

Get-Process

Get-Process | Sort-Object WS -Descending | Select -First 10

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 4: Services

Aim:

Learn the associated PowerShell cmdlets.

Commands:

Get-Service

Get-Service | Where-Object Status -eq Running

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 5: File System

Aim:

Learn the associated PowerShell cmdlets.

Commands:

Get-ChildItem

New-Item LabFiles -ItemType Directory

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 6: Files

Aim:

Learn the associated PowerShell cmdlets.

Commands:

```
New-Item Test.txt -ItemType File  
Rename-Item Test.txt Sample.txt
```

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 7: Read/Write Files

Aim:

Learn the associated PowerShell cmdlets.

Commands:

```
"Hello"|Out-File sample.txt  
Get-Content sample.txt
```

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 8: Environment

Aim:

Learn the associated PowerShell cmdlets.

Commands:

```
Get-ChildItem Env:  
$env:PATH
```

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.

2. What output did you observe?

Experiment 9: Networking

Aim:

Learn the associated PowerShell cmdlets.

Commands:

```
Get-NetIPAddress
```

```
Test-Connection google.com -Count 2
```

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?

Experiment 10: Reports

Aim:

Learn the associated PowerShell cmdlets.

Commands:

```
Get-Process | Export-Csv processes.csv -NoTypeInfo
```

Exercise:

Execute the commands, observe the output, and write your observations.

Viva Questions:

1. Explain the cmdlets used.
2. What output did you observe?